

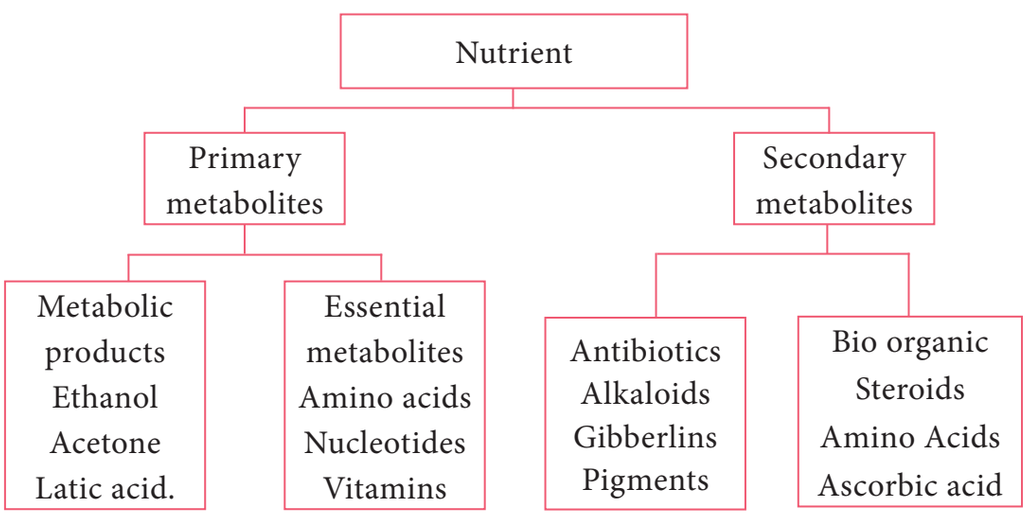
Figure 6.1: Bubble formation in grape juice fermentation

6.1 Industrially Important Microbes and their Products

Microorganisms have the powerful capacity to produce numerous products, during their life cycle. Flowchart 6.1 shows the production of valuable metabolic products during the growth of microorganisms on a suitable medium under controlled environmental conditions. Microbial products are often classified as primary and secondary metabolites.

Primary metabolites consist of compounds related to the synthesis of microbial cells in the growth phase. Primary metabolites such as amino acids, vitamins, enzymes, organic acids and nitrogenous bases are produced by wide variety of microorganisms. These primary metabolites are essential for the growth of microorganisms and they are produce during Logarithmic phase. Secondary metabolites do not play a role in development, growth and reproduction of microorganisms. They are produce at the end of growth phase near stationary phase. They usually accumulate during the period of nutrient limitation or waste product accumulation that follows the exponential phase. These compounds have no direct relationship to the synthesis of cell materials and normal growth. They are the end products of the primary metabolism. Products such as steroids, alkaloids, antibiotics are secondary metabolites.

Excessive production of the primary and secondary metabolites produced by the microorganisms are useful in the large scale in industrial production. Unlike primary metabolites, secondary metabolites are produced in small quantities and their extraction is difficult (Figure 6.2).



Flowchart 6.1: Various metabolites produced in Industrial fermentation

TN_GOVT_XII_Micro_Biology_CH06.indd 86 27-02-2019 14:46:00